Experiment 1
Code

```
name, age, department = input("Enter Name age department ").split()
age = int(age)
marks = list(map(int, input("Enter the 5 Subject mark ").split()))
total = sum(marks)
average = total / len(marks)
highest = max(marks)
lowest = min(marks)
print("\nStudent details")
print(f"Name : {name}")
print(f"Age : {age}")
print(f"Mark : {marks}")
print(f"Total : {total}")
print(f"Average : {average}")
print(f"Department : {department}")
print(f"Highest : {highest}")
print(f"Lowest : {lowest}")
print(f"Data Type of mark : {type(marks)}")
print(f"Number of Subject : {len(marks)}")
print(f"Rounded Average : {round(average)}")
grade = input("\nEnter the Grade (A/B/C/D) ")
print(f"ASCII value of {grade} is {ord(grade)}")
ascii_value = int(input("Enter the ASCII Value "))
print(f"Character : {chr(ascii_value)}")
print("\nSubject numbers")
for i in range(1, 6):
    print(i)
if age >= 18:
    print("Eligible")
else:
    print("Not Eligible")
```

output:

```
Enter Name age department Alice
20 CS Enter the 5 Subject mark 85
90 78 92 88

 Student
details
Name :
Alice
Age : 20
Mark : [85, 90, 78, 92, 88]
total : 433
Average
: 86.6
Departme
nt : CS
Highest
: 92
```

```
Lowest : 78

Data Type of mark <class
'list'> Number of
Subject 5
Rounded Average : 87

 Enter the Grade
(A/B/C/D) A ASCII value
of A is 65
Enter the ASCII Value 66
Character : B

 Subject
numbers
1
2
3
4
5

 Eligible
```

Experiment2

Code

```
name, age, membership = input("Enter Name Age Membership: ").split()
age = int(age)
prices = list(map(float, input("Enter price of 5 products:
").split()))
total = sum(prices)
average = total / len(prices)
highest = max(prices)
lowest = min(prices)
print("\nCustomer Details")
print(f"Name : {name}")
print(f"Age : {age}")
print(f"Membership : {membership}")
print(f"Item Prices : {prices}")
print(f"Total Prices : {total}")
print(f"Average : {average}")
print(f"Highest : {highest}")
print(f"Lowest : {lowest}")
print(f"Data Type of Price List : {type(prices)}")
print(f"Number of Items Purchased : {len(prices)}")
print(f"Rounded Average : {round(average, 2)}")
coupon = input("\nEnter Coupon Code Character: ")
print(f"ASCII Value : {ord(coupon)}")
ascii_value = int(input("Enter ASCII Value: "))
print(f"Character : {chr(ascii_value)}")
print("\nItem Numbers")
for i in range(1, 6):
    print(i)
```

```
mem = {"Gold": 50, "Platinum": 40, "Silver": 30}
discount = mem[membership.capitalize()]
if age >= 60:
    print("\nCustomer is senior citizen")
else:
    print("\nCustomer is not senior citizen")
print("\nItem Prices")
for price in prices:
    print(price)
while True:
    print("\n----- MENU ----------")
    print("1. Show Bill")
    print("2. Show Discount")
    print("3. Exit")
    choice = int(input("Enter choice: "))
    if choice == 1:
        print(f"Total Bill : {round(total, 2)}")
        print(f"Average : {round(average, 2)}")
    elif choice == 2:
        print(f"Discount : {discount}")
        final_bill = total - discount
        print(f"Final Bill : {final_bill}")

    elif choice == 3:
        break
    else:
        print("Invalid Choice")
print("\n===== FINAL BILL =====")
print(f"Customer : {name}")
print(f"Membership : {membership}")
print(f"Total Bill : {total}")
print(f"Discount : {discount}")
print(f"Amount to Pay : {total - discount}")
print("Thank you! Visit Again.")
```

output:

```
Enter Name Age Membership John 25
Gold Enter price of 5 product 100
200 150 50 250

 Customer
Details
Name :
John
Age : 25
Membershi
p : Gold
Item Prices : [100.0, 200.0, 150.0, 50.0, 250.0]
Total prices :
750.0 Average
: 150.0
```

```
Highest : 250.0
Lowest : 50.0

Data Type of Price List : <class
'list'> Number of Items purchased
: 5
Rounded Average : 150.0

 Enter Coupon Code
Character : A ASCII
Value : 65
Enter ASCII
Value : 66
Character : B

 Ite
m
Numb
er 1
2
3
4
5

 Customer is not senior citizen
 Item
prices
100.0
200.0
150.0
50.0
250.0

      MENU
1. Show Bill
2. Show Discount
3. Exit
Enter choice : 3
```

Experiment3

Code

```
d = {1: 1001, 2: 2002}
ba = [2000, 4000]
las = {1: "", 2: ""}
while True:
    ac = int(input("Account number : "))
    if ac in d:
        b = int(input("Pin : "))
        if d[ac] == b:
            print("Account Verified")
            break
        else:
```

```
            print("Wrong Pin")
            continue
    else:
        print("Wrong account number : ")
        continue
print("""1. View balance
2. Deposit money
3. Withdraw money
4. View last few transaction
5. Exit""")
while True:
    a = int(input("Enter the option (number) : "))
    if a == 1:
        print(ba[ac - 1])
    elif a == 2:
        am = int(input("Enter amount to deposit : "))
        ba[ac - 1] = ba[ac - 1] + am
        las[ac] = las[ac] + f" deposited {am}"
    elif a == 3:
        am = int(input("Enter amount to withdraw : "))
        if ba[ac - 1] >= am:
            ba[ac - 1] = ba[ac - 1] - am
            las[ac] = las[ac] + f" withdrew {am}"
        else:
            print("Insufficient balance")
    elif a == 4:
        print(las[ac])
    elif a == 5:
        print("Thank you")
        exit()

output:
Enter the Account Number : 1
Enter the Pin : 1001
Account Verified

1. View balance
2. Deposit money
3. Withdraw money
4. View last few transaction
5. Exit

Enter the option (number) : 1
2000

Enter the option (number) : 2
Enter the amount to deposit : 100

Enter the option (number) : 3
Enter the amount to withdraw : 100

Enter the option (number) : 4
```

```
deposited 100 withdrew 100

Enter the option (number) : 1
2000

Enter the option (number) : 5
Thank you.
```

Experiment4

Code

```
Name = input('Name : ').strip().title()
Email = input('Email : ').strip().lower()
Username = input('Username : ').strip().lower()
Mobile = input("Mobile : ").strip().lower()
print("@ in mail : ", "@" in Email)
print("@ position : ", Email.find("@"))
print("Slicing : ", (Email[:-10], Email[-10:]))
print("Split : ", Email.split("@"))
print(".Com : ", Email.endswith(".com"))
print("Username alphanumeric : ", Username.isalnum())
print("Mobile digits : ", Mobile.isdigit())
Username = Username.replace("_", " ")
print("a count ", Name.lower().count("a"))
rev = Username[::-1]
Userid = rev[:3] + Mobile[-4:]
print(f'''
Name : {Name}
Mobile : {Mobile}
Email : {Email}
Username : {rev}
UserID : {Userid}
Username length : {len(Username)}
First 3 : {Username[:3]}
Last 4 : {Username[-4:]}
''')
```

Output

```
Name : ab
Email : ab@gmail.com
Username : ab_1234
Mobile Number : 9876543210

@ in email : True
@ Position : 2
Slicing : ('ab', '@gmail.com')
Split : ['ab', 'gmail.com']
.Com : True
Username alphanumeric : False
Mobile digits : True
a count : 1

Name : Ab
```

```
Mobile : 9876543210
Email : ab@gmail.com
Username : 4321 a b
User ID : 4323210
Username length : 7
First 3 : ab_
Last 4 : 1234
```

Experiment5

Code

```
def check_data_usage(total_data, data_used):
    remaining_data = total_data - data_used
    if remaining_data <= 0:
        message = "Data limit exceeded, please recharge"
    elif remaining_data <= 0.2 * total_data:
        message = "Data low, consider recharging"
    else:
        message = "Data usage is normal"
    return remaining_data, message
total_data = float(input("Enter your total data plan (in GB): "))
data_used = float(input("Enter the data used (in GB): "))
remaining_data, message = check_data_usage(total_data, data_used)
print("------------ MOBILE DATA USAGE REPORT ------------")
print(f"Data Plan : {total_data:.1f} GB")
print(f"Data Used : {data_used:.1f} GB")
if remaining_data < 1 and remaining_data >= 0:
    remaining_data_mb = remaining_data * 1024
    print(f"Remaining Data : {remaining_data_mb:.2f} MB")
else:
    print(f"Remaining Data : {remaining_data:.2f} GB")
print(f"Alert Message: {message}")
print("-------------------------------------------------")
```

output

```
Enter Your total data plan (in GB): 10
Enter the data used (in GB): 2

------------ MOBILE DATA USAGE REPORT ------------
Data plan : 10.0 GB
Data used : 2.0 GB
Remaining Data : 8.00 GB
Alert message: Data usage is normal
```

Experiment6
Code

```
def display_title():
    print("=" * 55)
    print(" Railway Ticket Booking System")
    print("=" * 55)
    print("Welcome! Enter Passenger and Journey details.\n")
def display_passenger(**kwargs):
    print("\nPassenger Information")

    for key, value in kwargs.items():
        print(f"{key.title()}: {value}")
def calculate_basic_fare(distance, fare_per_km=2.50):
    return distance * fare_per_km
def age_discount(age, discount_rate=0.10):
    if age < 12:
        return 0.5
    elif age >= 60:
        return 0.3
    return discount_rate
def calculate_discount(fare, rate):
    return fare * rate
def calculate_charges(*args):
    return sum(args)
def
calculate_operations(amount,discount=0,tax=0,service=0,convenience=0):
    return {
        "Discount": amount * discount,
        "Tax": amount * tax,
        "Service Charge": amount * service,
        "Convenience Fee": amount * convenience}
def final_ticket(**kwargs):
    print("\n" + "=" * 55)
    print(" FINAL TICKET")
    print("=" * 55)
    for key, value in kwargs.items():
        print(f"{key:<22}: {value}")
    print("=" * 55)
def process_passengers(passengers, index=0, total=0):
    if index == len(passengers):
        return total
    passenger = passengers[index]
    basic_fare = calculate_basic_fare(passenger["distance"])
    rate = age_discount(passenger["age"])
    discount = calculate_discount(basic_fare, rate)
    food = passenger["food_charge"]
    baggage = passenger["baggage_charge"]
    service = passenger["service_charge"]
    fare_after_discount = basic_fare - discount
    extra = calculate_charges(food, baggage, service)
    final_amount = fare_after_discount + extra
```

```
    passenger['basic_fare'] = basic_fare
    passenger['discount'] = discount
    passenger['additional_charges'] = extra
    passenger['final_amount'] = final_amount
    return process_passengers(
        passengers,
        index + 1,
        total + final_amount
    )
display_title()
n = int(input("Enter the number of passengers: "))
passengers = []
for i in range(n):
    print(f"\n[1] Passenger {i + 1}")
    name = input("Name: ")
    age = int(input("Age: "))
    gender = input("Gender: ")
    source = input("Source station: ")
    destination = input("Destination Station: ")
    distance = float(input("Distance: "))
```

output:

```
============================================================
 Railway Ticket Booking System
============================================================
Welcome! Enter Passenger and Journey details.

Enter the number of passengers: 2

[1] Passenger 1
Name: a
Age: 10
Gender: male
Source station: b
Destination Station: c
Distance: 10

[1] Passenger 2
Name: d
Age: 30
Gender: female
Source station: e
Destination Station: f
Distance: 10
```